In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
%matplotlib inline
```

In [2]:
```python
df = pd.read_csv("matches.csv")

df.head()
```

Out:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result	dl_applied	winner	win_by_runs	wi
0	1	2017	Hyderabad	5/4/2017	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0
1	2	2017	Pune	6/4/2017	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7
2	3	2017	Rajkot	7/4/2017	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10
3	4	2017	Indore	8/4/2017	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6
4	5	2017	Bangalore	8/4/2017	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	normal	0	Royal Challengers Bangalore	15	0

In [3]:
```python
print("Shape:", df.shape)

df.info()

df.describe()

df.isnull().sum()
```

Out:
```
Shape: (636, 18)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 636 entries, 0 to 635
Data columns (total 18 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   id               636 non-null    int64 
 1   season           636 non-null    int64 
 2   city             629 non-null    object
 3   date             636 non-null    object
 4   team1            636 non-null    object
 5   team2            636 non-null    object
 6   toss_winner      636 non-null    object
 7   toss_decision    636 non-null    object
 8   result           636 non-null    object
 9   dl_applied       636 non-null    int64 
 10  winner           633 non-null    object
```

```
 11  win_by_runs      636 non-null    int64
 12  win_by_wickets   636 non-null    int64
 13  player_of_match  633 non-null    object
 14  venue            636 non-null    object
 15  umpire1          635 non-null    object
 16  umpire2          635 non-null    object
 17  umpire3          0 non-null      float64
dtypes: float64(1), int64(5), object(12)
memory usage: 89.6+ KB
```

```
id                   0
season               0
city                 7
date                 0
team1                0
team2                0
toss_winner          0
toss_decision        0
result               0
dl_applied           0
winner               3
win_by_runs          0
win_by_wickets       0
player_of_match      3
venue                0
umpire1              1
umpire2              1
umpire3            636
dtype: int64
```

In [4]:

```python
winning_teams = df['winner'].value_counts()

print(winning_teams)
```

Out:

```
winner
Mumbai Indians                 92
Chennai Super Kings            79
Kolkata Knight Riders          77
Royal Challengers Bangalore    73
Kings XI Punjab                70
Rajasthan Royals               63
Delhi Daredevils               62
Sunrisers Hyderabad            42
Deccan Chargers                29
Gujarat Lions                  13
Pune Warriors                  12
Rising Pune Supergiant         10
Kochi Tuskers Kerala            6
Rising Pune Supergiants         5
Name: count, dtype: int64
```

In [5]:

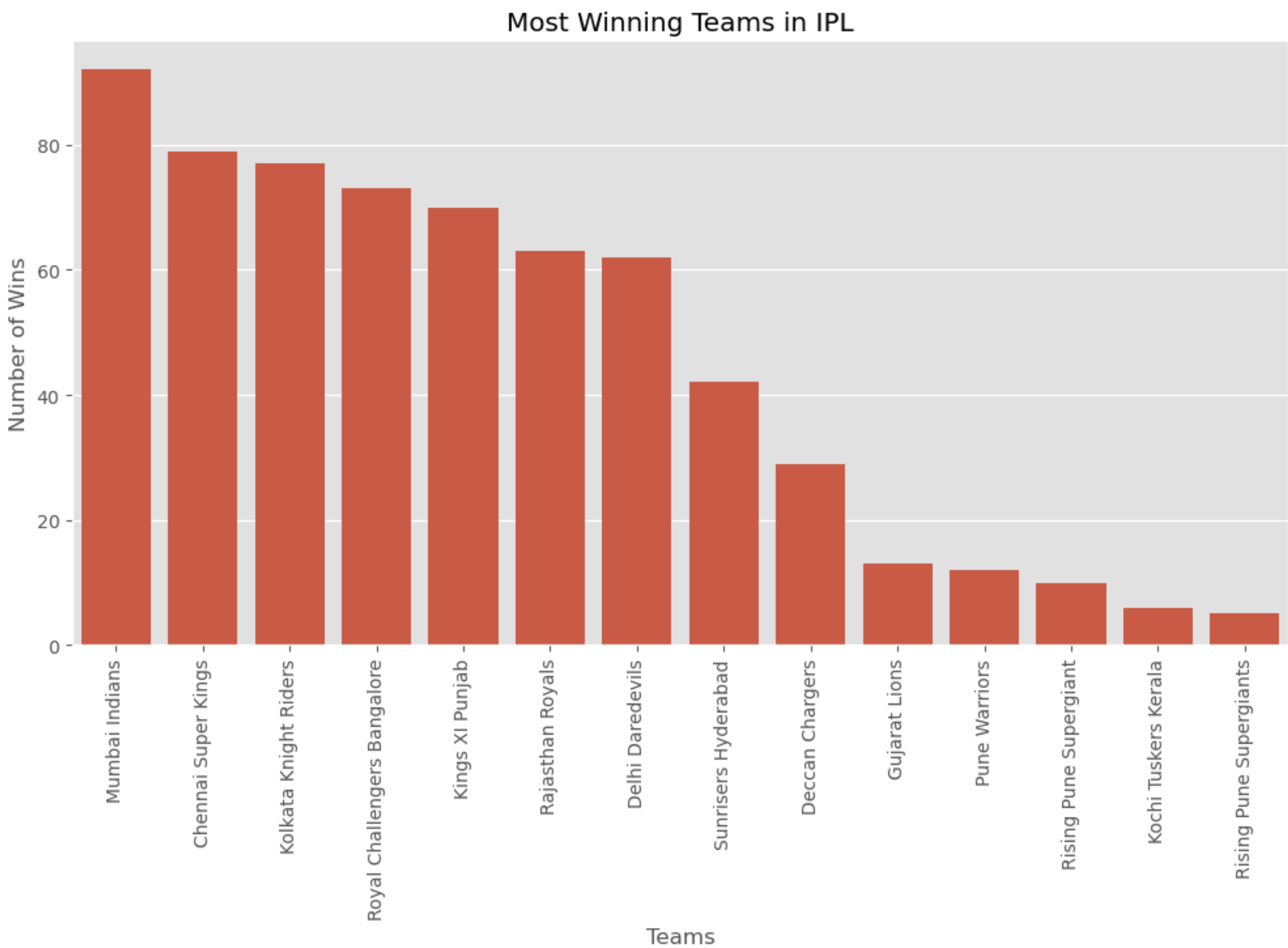
```python
plt.figure(figsize=(12,6))

sns.barplot(
    x=winning_teams.index,
    y=winning_teams.values,
)

plt.xticks(rotation=90)
plt.title("Most Winning Teams in IPL")
plt.xlabel("Teams")
```

```
plt.ylabel("Number of Wins")
plt.show()
```

Out:



In [6]:

```
print("Most Successful Team:", winning_teams.idxmax())
print("Total Wins:", winning_teams.max())
```

Out:

```
Most Successful Team: Mumbai Indians
Total Wins: 92
```

In [7]:

```
import os

print(os.getcwd())  # Current folder

print(os.listdir()) # Files in folder
```

Out:

```
C:\Users\USER\OneDrive\Desktop\IPL DATASET
['.ipynb_checkpoints', 'matches.csv', 'Untitled.ipynb', 'Untitled1.ipynb']
```

In [8]:

```
df.columns
```

Out:

```
Index(['id', 'season', 'city', 'date', 'team1', 'team2', 'toss_winner',
       'toss_decision', 'result', 'dl_applied', 'winner', 'win_by_runs',
       'win_by_wickets', 'player_of_match', 'venue', 'umpire1', 'umpire2',
       'umpire3'],
      dtype='object')
```

In [9]:

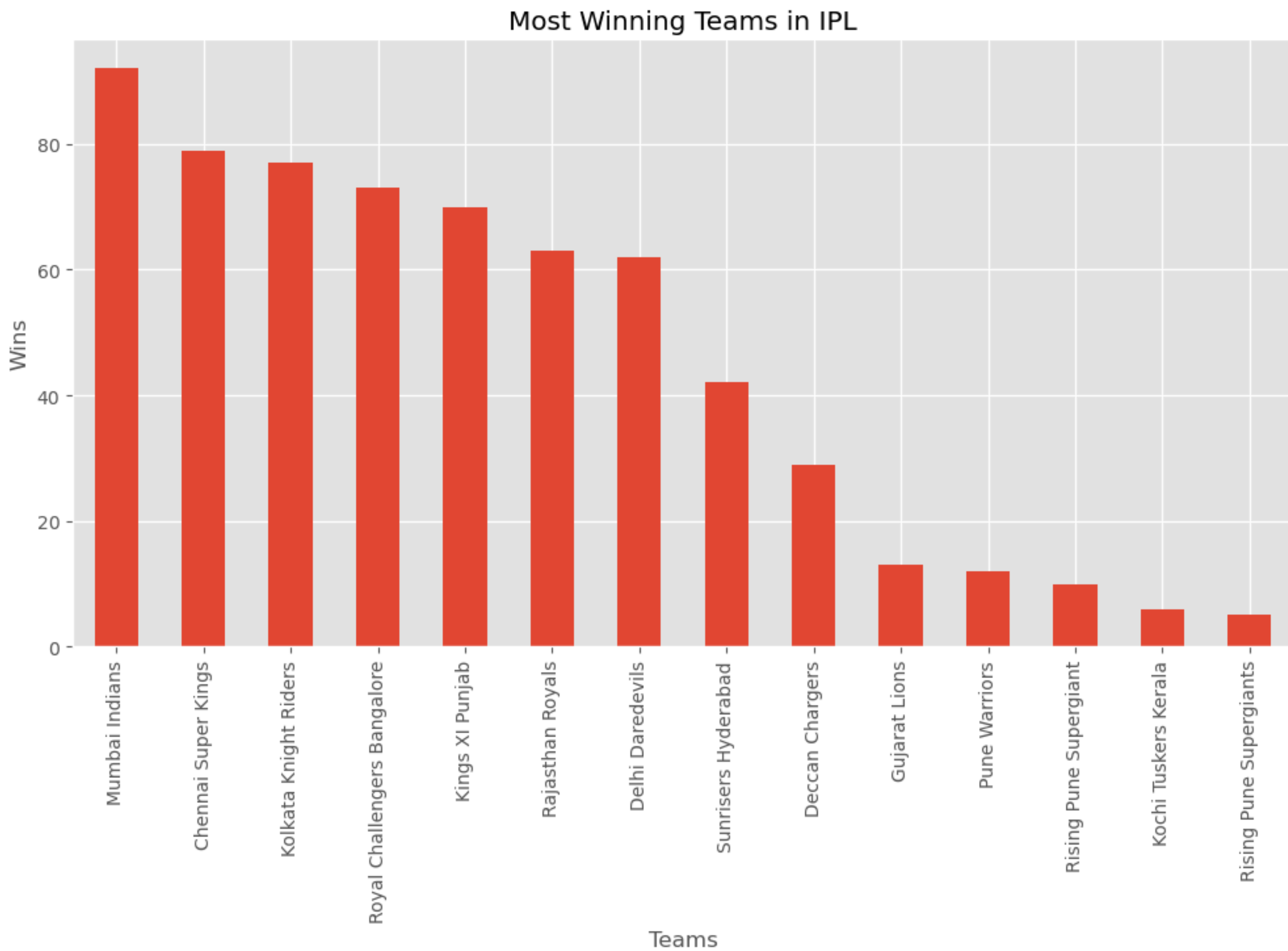
```
winning_teams = df['winner'].value_counts()

plt.figure(figsize=(12,6))
winning_teams.plot(kind='bar')

plt.title('Most Winning Teams in IPL')
plt.xlabel('Teams')
plt.ylabel('Wins')
plt.show()

print("Most Winning Team:", winning_teams.idxmax())
```

Out:




```
Most Winning Team: Mumbai Indians
```

In [10]:

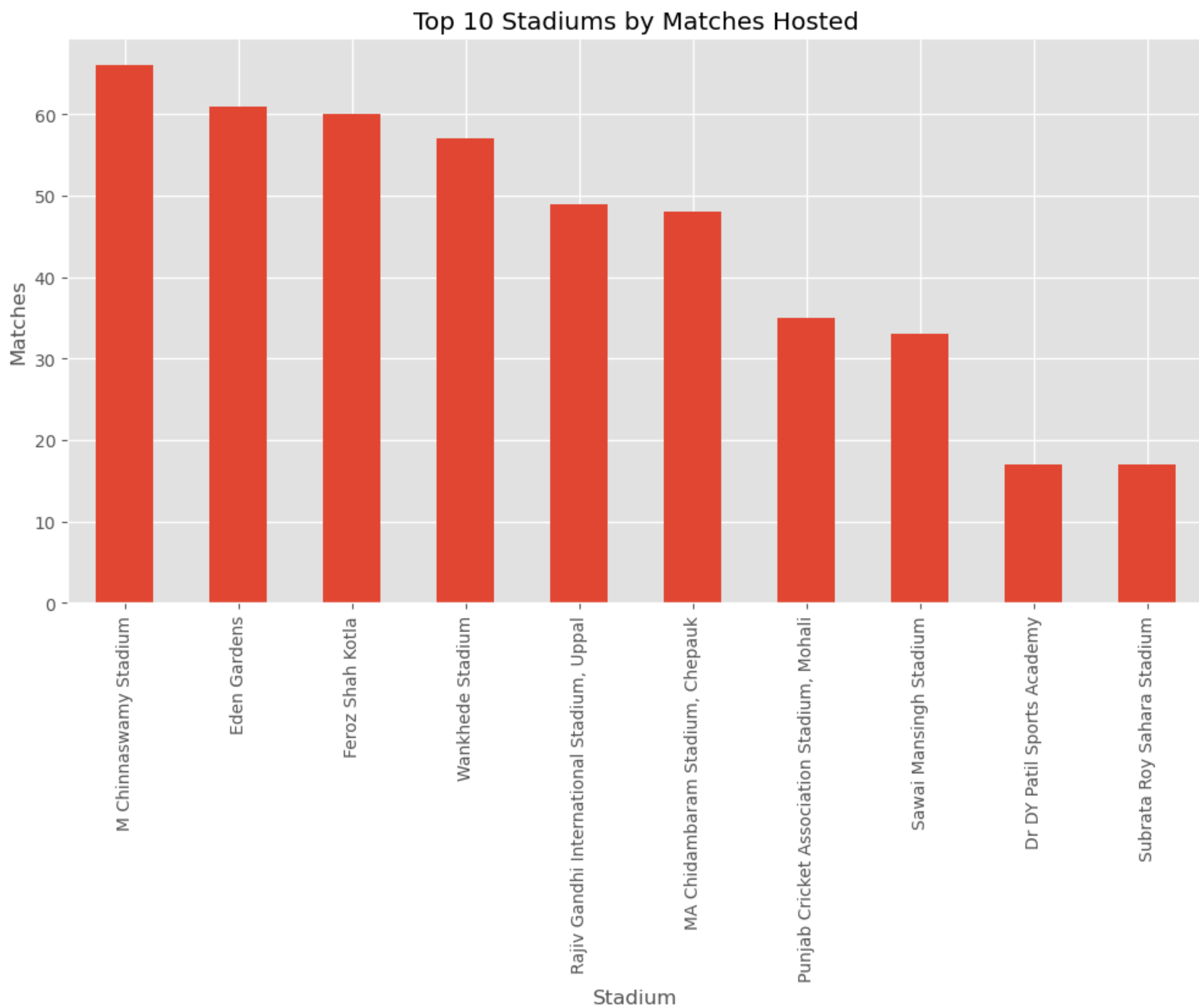
```
stadium_matches = df['venue'].value_counts()

plt.figure(figsize=(12,6))
stadium_matches.head(10).plot(kind='bar')

plt.title('Top 10 Stadiums by Matches Hosted')
```

```
plt.xlabel('Stadium')
plt.ylabel('Matches')
plt.show()
```

Out:



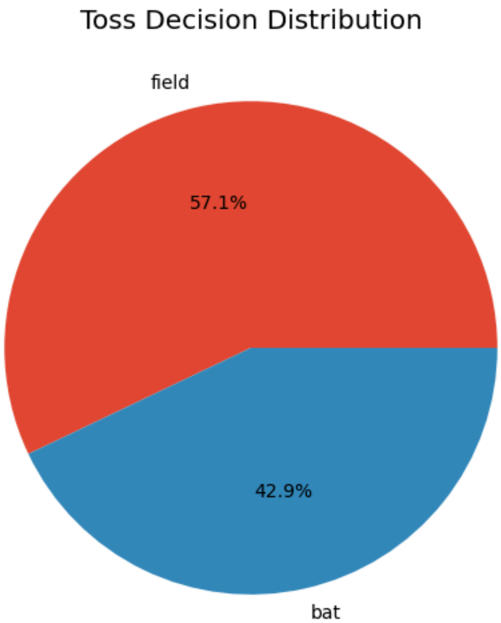
In [11]:

```
toss = df['toss_decision'].value_counts()

plt.figure(figsize=(6,6))
plt.pie(
    toss.values,
    labels=toss.index,
    autopct='%1.1f%%'
)

plt.title('Toss Decision Distribution')
plt.show()
```

Out:



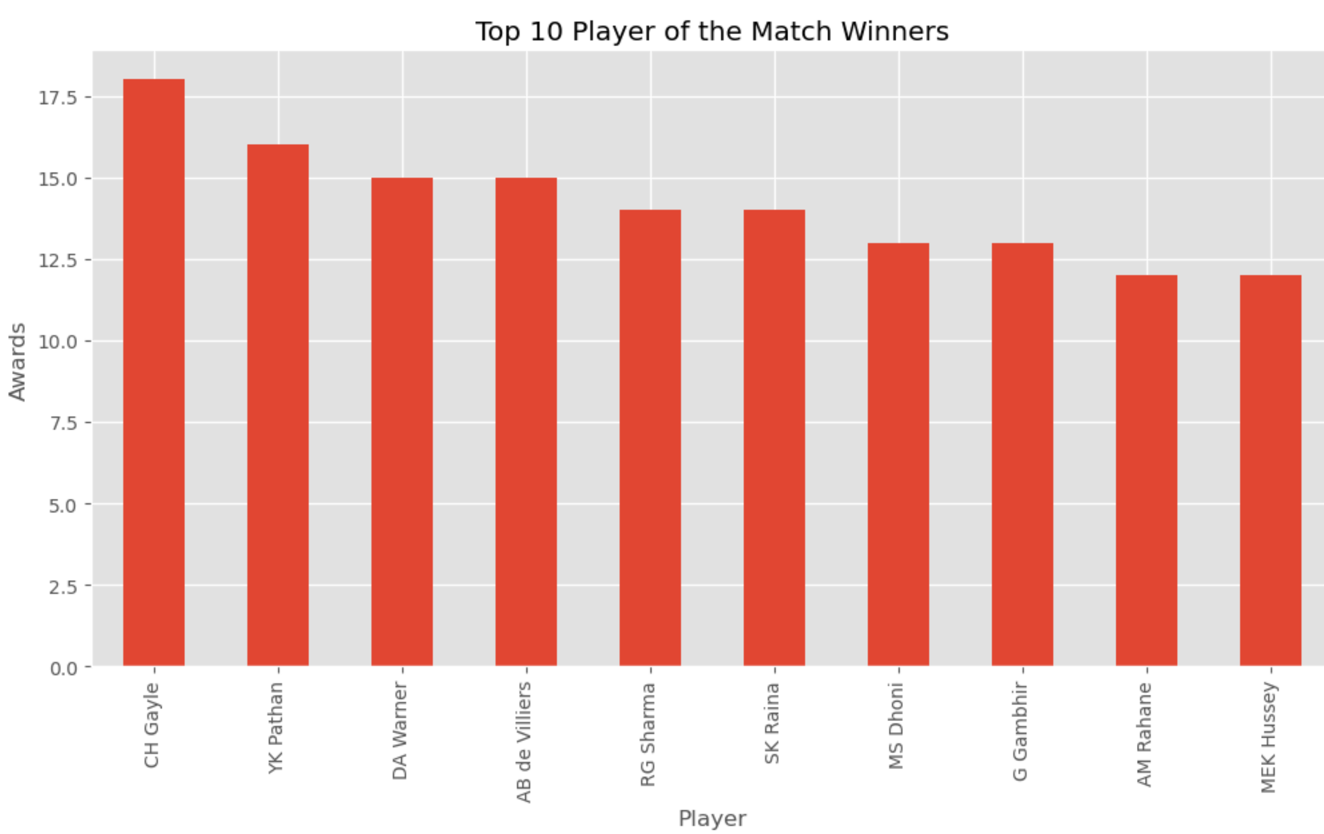
In [12]:

```
pom = df['player_of_match'].value_counts().head(10)

plt.figure(figsize=(12,6))
pom.plot(kind='bar')

plt.title('Top 10 Player of the Match Winners')
plt.xlabel('Player')
plt.ylabel('Awards')
plt.show()
```

Out:



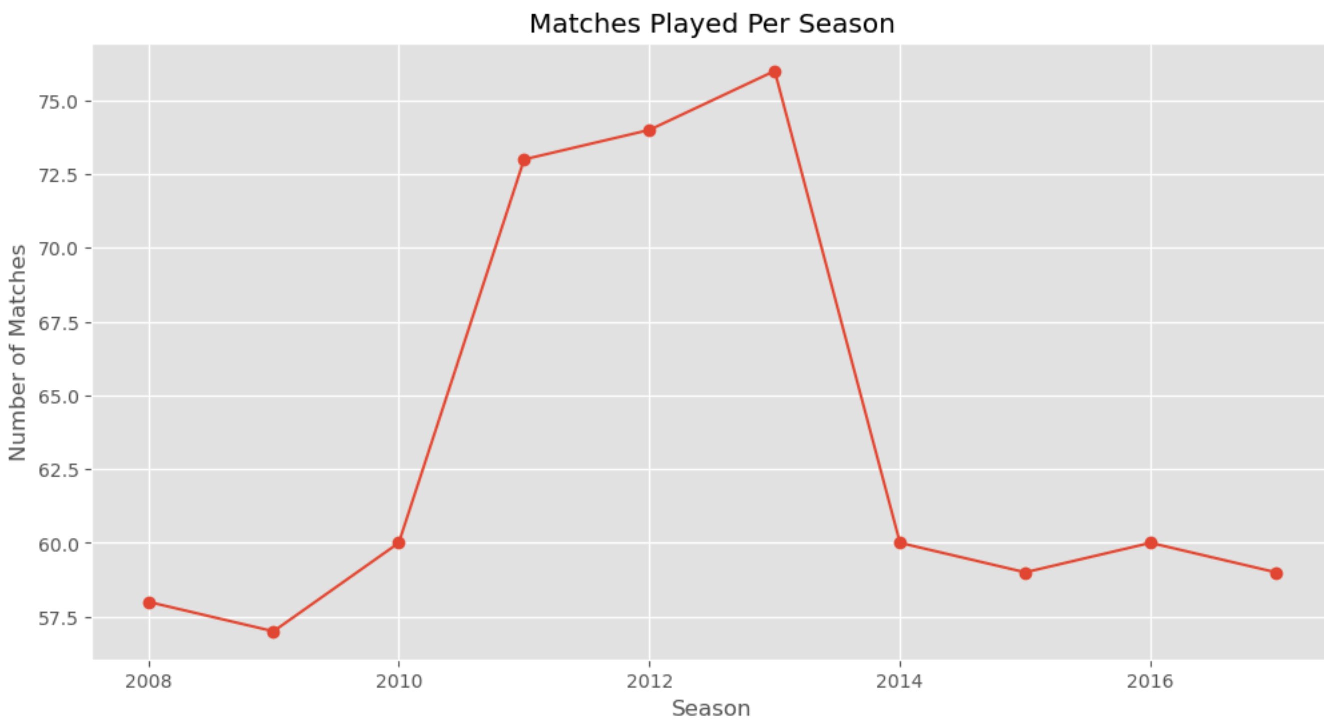
In [13]:

```
season_matches = df['season'].value_counts().sort_index()

plt.figure(figsize=(12,6))
season_matches.plot(marker='o')

plt.title('Matches Played Per Season')
plt.xlabel('Season')
plt.ylabel('Number of Matches')
plt.show()
```

Out:



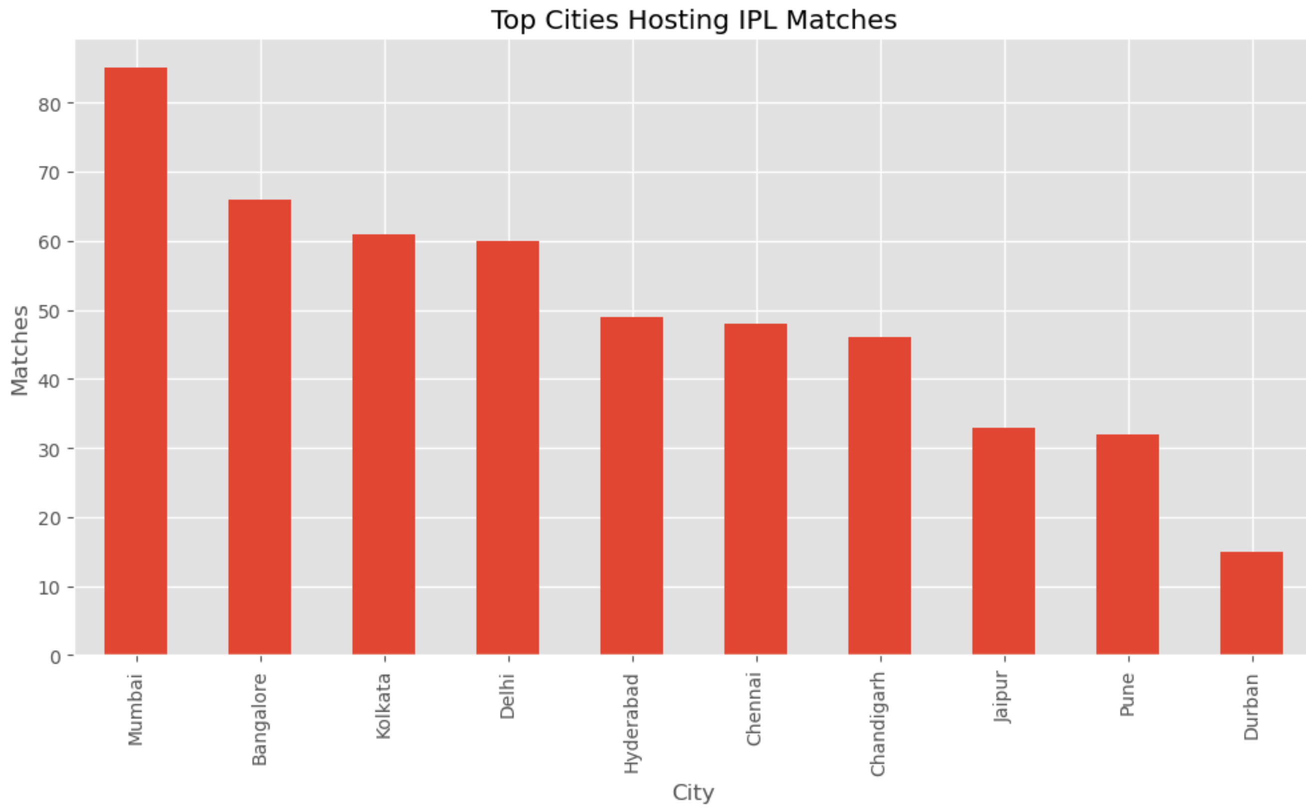
In [14]:

```
cities = df['city'].value_counts().head(10)

plt.figure(figsize=(12,6))
cities.plot(kind='bar')

plt.title('Top Cities Hosting IPL Matches')
plt.xlabel('City')
plt.ylabel('Matches')
plt.show()
```

Out:



In [15]:

```
print("===== IPL EDA INSIGHTS =====")

print("Most Winning Team:", df['winner'].value_counts().idxmax())

print("Most Used Stadium:", df['venue'].value_counts().idxmax())

print("Most Player of Match Awards:",
      df['player_of_match'].value_counts().idxmax())

print("Most IPL Matches Hosted City:",
      df['city'].value_counts().idxmax())
```

Out:

```
===== IPL EDA INSIGHTS =====
Most Winning Team: Mumbai Indians
Most Used Stadium: M Chinnaswamy Stadium
Most Player of Match Awards: CH Gayle
Most IPL Matches Hosted City: Mumbai
```

In [16]:

```
# Check missing values

df.isnull().sum()
```

Out:

```
id                  0
season              0
city                7
date                0
team1               0
team2               0
toss_winner         0
toss_decision       0
result              0
```

```
dl_applied           0
winner               3
win_by_runs          0
win_by_wickets       0
player_of_match      3
venue                0
umpire1              1
umpire2              1
umpire3            636
dtype: int64
```

In [17]:

```
# Remove rows where winner is missing

df = df.dropna(subset=['winner'])

print(df.shape)
```

Out:

```
(633, 18)
```

In [18]:

```
# Check duplicates

print("Duplicate Rows:", df.duplicated().sum())

# Remove duplicates if any

df = df.drop_duplicates()
```

Out:

```
Duplicate Rows: 0
```

In [19]:

```
print("Number of Matches:", len(df))
print("Number of Seasons:", df['season'].nunique())
print("Number of Teams:", len(pd.unique(df[['team1','team2']].values.ravel())))
print("Number of Cities:", df['city'].nunique())
```

Out:

```
Number of Matches: 633
Number of Seasons: 10
Number of Teams: 14
Number of Cities: 30
```

In [20]:

```
matches_played = pd.concat([df['team1'], df['team2']]).value_counts()

matches_won = df['winner'].value_counts()

win_percentage = (matches_won / matches_played * 100).sort_values(ascending=False)

win_percentage = win_percentage.dropna()

win_percentage.head(10)
```

Out:

```
Rising Pune Supergiant         62.500000
Chennai Super Kings            60.305344
Mumbai Indians                 58.598726
Sunrisers Hyderabad            55.263158
```

```
Rajasthan Royals               53.846154
Kolkata Knight Riders          52.027027
Royal Challengers Bangalore    48.666667
Kings XI Punjab                47.297297
Gujarat Lions                  43.333333
Kochi Tuskers Kerala           42.857143
Name: count, dtype: float64
```

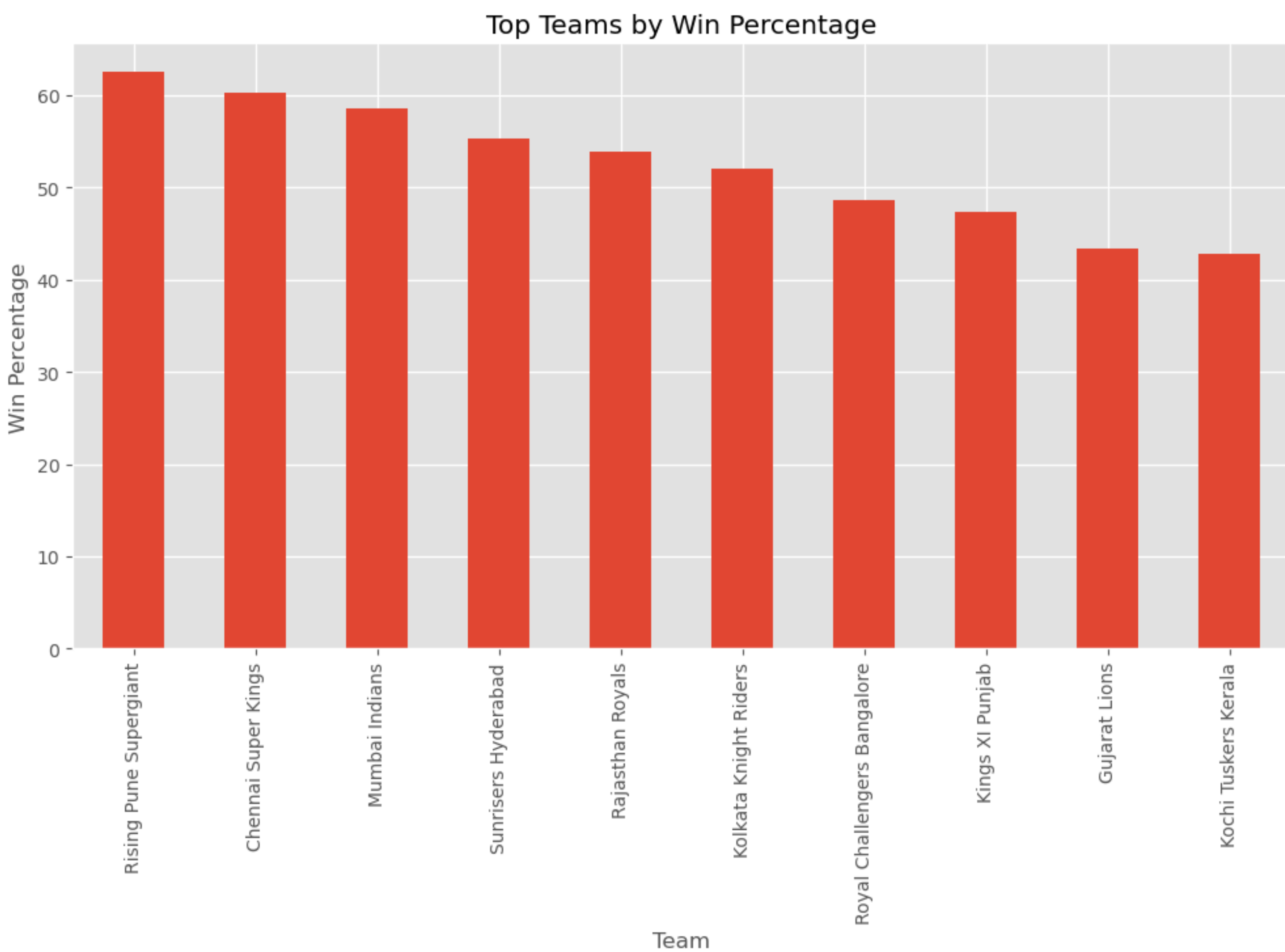
In [21]:

```python
plt.figure(figsize=(12,6))

win_percentage.head(10).plot(kind='bar')

plt.title('Top Teams by Win Percentage')
plt.xlabel('Team')
plt.ylabel('Win Percentage')
plt.show()
```

Out:



In [22]:

```python
umpires = pd.concat([df['umpire1'], df['umpire2']])

top_umpires = umpires.value_counts().head(10)

plt.figure(figsize=(12,6))

top_umpires.plot(kind='bar')

plt.title('Top Umpires in IPL')
plt.xlabel('Umpire')
```

```
plt.ylabel('Matches Officiated')
plt.show()
```